

Android Notification

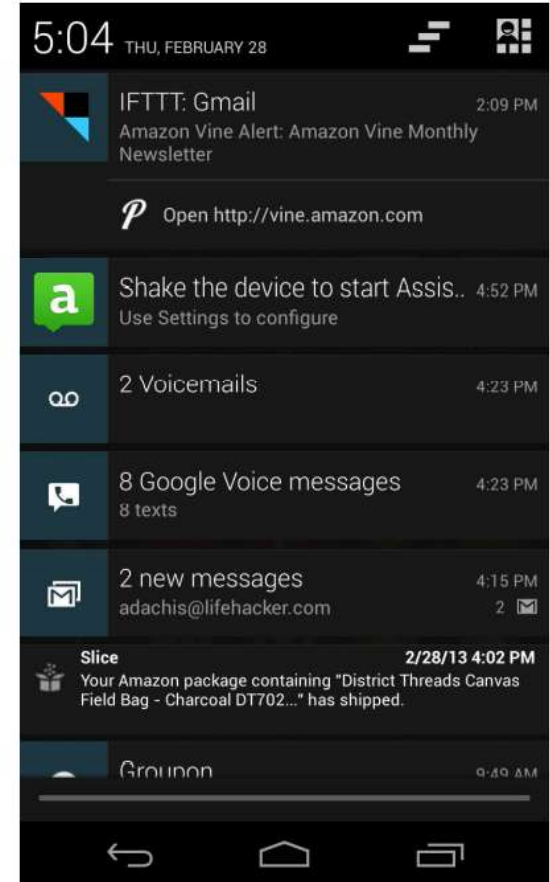
Contents

- **What are notifications?**
- **Notification channels**
- **Creating a notification channel**
- **Creating notifications**
- **Handling notification actions**



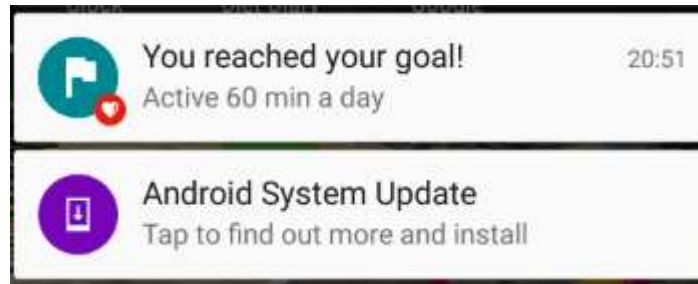
What are notifications

- Notification: A message displayed to the user outside of any app's UI in a top notification drawer area.
 - used to indicate system events, status of service tasks, etc.



What are notifications

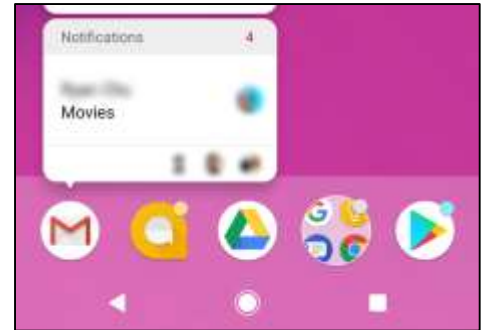
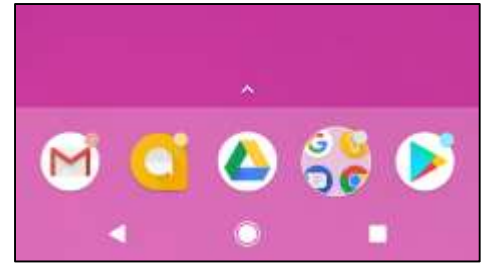
- **Notifications can have:**
 - icons (small, large)
 - a title
 - a detailed description
 - one or more associated actions that will occur when clicked
 - ..



App icon badge

- Available only on the devices running Android 8.0 (API level 26) and higher.

- New notifications are displayed as a colored "badge" (also known as a "notification dot") on the app icon.
- Users can long-press on an app icon to see the notifications for that app. Similar to the notification drawer.



Notification channels

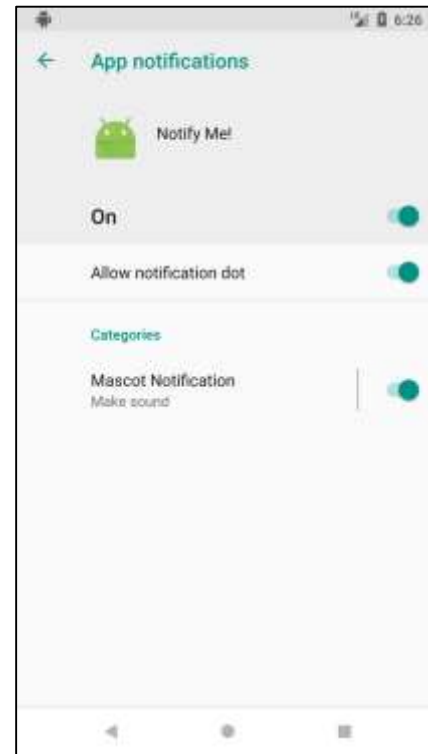
- Used to create a user-customizable channel for each type of notification to be displayed.
- More than one notification can be grouped in to a channel.
- Set notification behavior like sound, light, vibrate and so on, applied to all the notifications in that channel.

Notification channels are mandatory

- [Notification channels](#) are introduced in Android 8.0 (API level 26)
- All notifications must be assigned to a channel starting from Android 8.0 (API level 26), else your notifications will not be displayed.
- For the apps targeting lower than Android 8.0 (API level 26), no need to implement notification channels.

Notification channels in Settings

- Notification channels appear as Categories under App notifications in the device Settings.



Create a Notification channel

- Notification channel instance is created using [NotificationChannel](#) constructor.
- You must specify:
 - An ID that's unique within your package.
 - User visible name of the channel.
 - The importance level for the channel.

```
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {  
    NotificationChannel notificationChannel =  
        new NotificationChannel(CHANNEL_ID, "Mascot Notification",  
        NotificationManager.IMPORTANCE_DEFAULT);  
}
```

Importance level

- Available in Android 8.0 (API level 26) and higher.
- Sets the intrusion level, like the sound and visibility for all notifications posted in the channel.
- Range from IMPORTANCE_NONE(0) to IMPORTANCE_HIGH(4).
- To support earlier versions of Android (Lower than API level 26), set the priority.

Notification priority

- Determines how the system displays the notification with respect to other notifications, in Android version Lower than API level 26.
- Set using the [setPriority\(\)](#) method for each notification.
- Range from [PRIORITY_MIN](#) to [PRIORITY_MAX](#).

```
setPriority(NotificationCompat.PRIORITY_HIGH)
```

Importance level and priority constants

User-visible importance level	Importance (Android 8.0 and higher)	Priority (Android 7.1 and lower)
Urgent Makes a sound and appears as a heads-up notification	<u>IMPORTANCE_HIGH</u>	<u>PRIORITY_HIGH</u> or <u>PRIORITY_MAX</u>
High Makes a sound	<u>IMPORTANCE_DEFAULT</u>	<u>PRIORITY_DEFAULT</u>
Medium No sound	<u>IMPORTANCE_LOW</u>	<u>PRIORITY_LOW</u>
Low No sound and doesn't appear in the status bar	<u>IMPORTANCE_MIN</u>	<u>PRIORITY_MIN</u>

Creating notifications

- Notification is created using [NotificationCompat.Builder](#) class.
- Pass the application context and notification channel ID to the constructor.
- The [NotificationCompat.Builder](#) constructor takes the notification channel ID, this is only used by Android 8.0 (API level 26) and higher, but this parameter is ignored by the older versions.

```
NotificationCompat.Builder mBuilder = new  
    NotificationCompat.Builder(this, CHANNEL_ID);
```

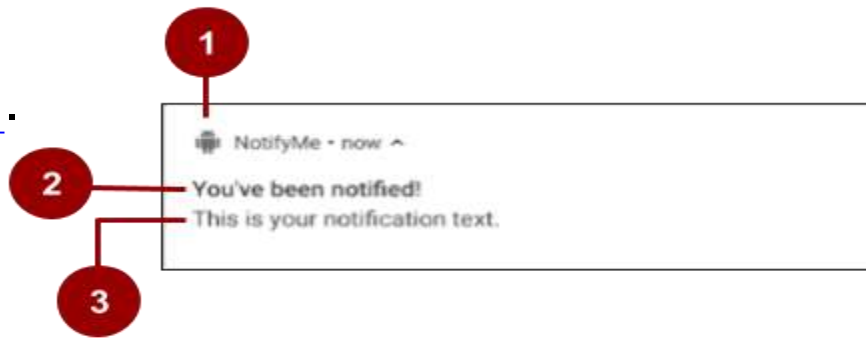
Setting notification contents

1. A small icon, set by [`setSmallIcon\(\)`](#).

This is the only content that's required.

1. A title, set by [`setContentTitle\(\)`](#).

2. The body text, set by [`setContentText\(\)`](#). This is the notification message.



Setting notification contents

```
NotificationCompat.Builder mBuilder =  
    new NotificationCompat.Builder(this, CHANNEL_ID)  
        .setSmallIcon(R.drawable.android_icon)  
        .setContentTitle("You've been notified!")  
        .setContentText("This is your notification text.");
```

Delivering notifications

```
Notification notification = mBuilder.build();
```

```
NotificationManager manager = (NotificationManager)  
    getSystemService(Context.NOTIFICATION_SERVICE);
```

```
manager.notify(ID, notification)
```


Updating notifications

1. Update a notification by changing and or adding some of its content.
2. Issue notification with updated parameters using builder.
3. Call `notify()` passing in the same notification ID.
 - If previous notification is still visible, system updates.
 - If previous notification has been dismissed, new notification is delivered.

Canceling notifications

Notifications remain visible until:

- User dismisses it by swiping or by using "Clear All".
- Calling [setAutoCancel\(\)](#) when creating the notification, removes it from the status bar when the user clicks on it.
- App calls `cancel()` or `cancelAll()` on `NotificationManager`.

```
mNotifyManager.cancel(NOTIFICATION_ID);
```

Handling notification actions

- Normally when the user clicks on a notification, an action should occur. (direct the user to a particular app / activity, etc.)
 - To achieve this, use an intent inside your notification.
 - Must wrap it inside a "pending intent" object.

```
Notification.Builder builder = ...;
```

```
Intent intent = new Intent(this, ActivityClassName.class);
```

```
intent.putExtra("key1", "value1");
```

```
...
```

```
PendingIntent pending = PendingIntent.getActivity(
```

```
this, 0, intent, 0);
```

```
builder.setContentIntent(pending);
```

```
Notification notification = builder.build();
```

Classwork

- When opening the app, if there is product in the cart
 - Show a notification on status bar to remind user about selected products
 - When clicking on the notification, open a Cart screen

References

- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>